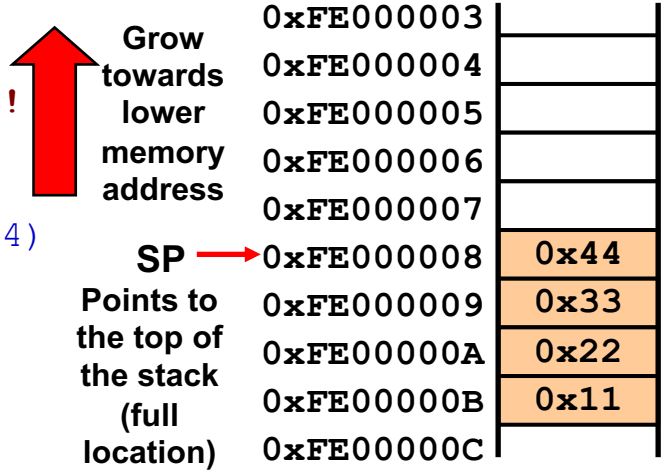
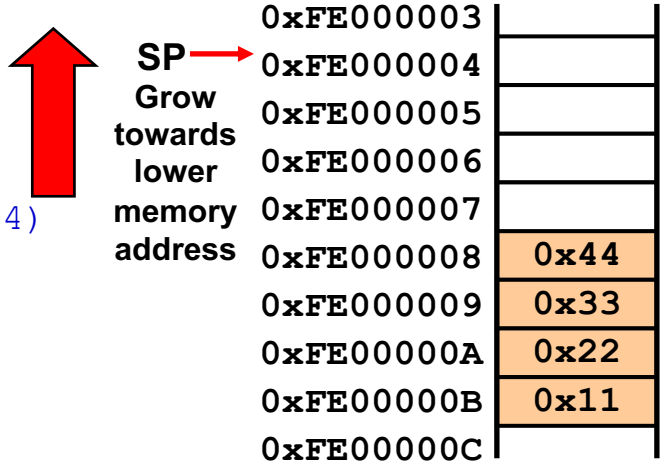
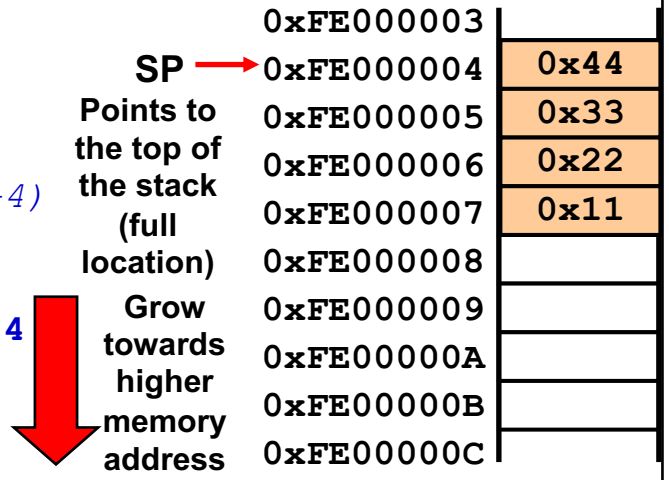
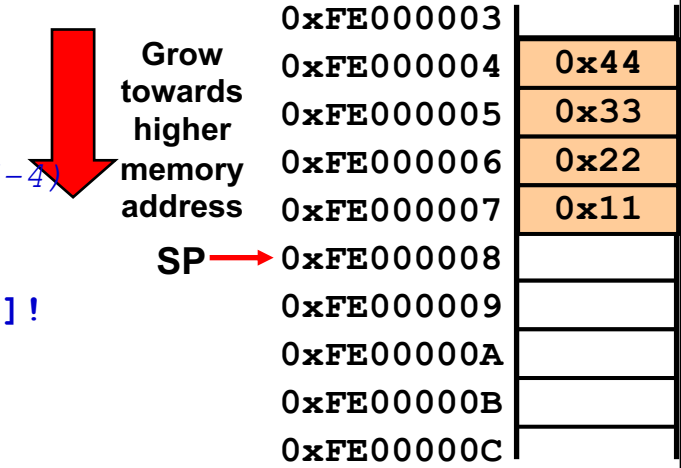
Stack Types Implementation

Asst. Prof. Mohamed M. Sabry Aly

©2024, SCSE/NTU

	Full	Empty
Descending	Push operation: *negative offset (-4)* *pre index (move pointer then store)* **STR Rx, [SP,#-4]!** Pop operation: *positive offset (+4)* *post index (load then move pointer)* **LDR Rx, [SP], #4** Grow towards lower memory address SP → Points to the top of the stack (full location)	Push operation: *negative offset (-4)* *post index (store then move pointer)* **STR Rx, [SP],#-4** Pop operation: *positive offset (+4)* *pre index (move pointer then load)* **LDR Rx, [SP,#4]!** Points to an empty position at the top of the stack SP → Grow towards lower memory address
Ascending	Push operation: *positive offset (+4)* *pre index (move pointer then store)* **STR Rx, [SP,#4]!** Pop operation: *negative offset (-4)* *post index (load then move pointer)* **LDR Rx, [SP], #-4** SP → Points to the top of the stack (full location) Grow towards higher memory address	Push operation: *positive offset (+4)* *post index (store then move pointer)* **STR Rx, [SP],#4** Pop operation: *negative offset (-4)* *pre index (move pointer then load)* **LDR Rx, [SP,#-4]!** Points to an empty position at the top of the stack Grow towards higher memory address SP →